Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 21)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

[4 marks]

0 1 . 2 Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

[2 marks]

0 1 . 3 Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

[2 marks]

0 1 . 4 Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

[2 marks]

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0 2 . 1 Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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0 2 . 2 Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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0 2 . 3 Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

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Advantage .....

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Disadvantage .....

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0 2 . 4 Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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0 3 Figure 1 compares the energy flows in a conventional gas-fired power station and a combined heat and power (CHP) station, each supplied with 100 units of fuel energy.

Figure 1

	Conventional power station	**CHP station**
Electricity output / units	40	35
Useful heat supplied to buildings / units	0	45
Waste heat lost / units	60	20

0 3 . 1 Calculate the overall efficiency of each station.

[2 marks]

Answer ..... %

0 3 . 2 Explain why the electrical efficiency of a CHP station is often deliberately kept below 40%.

[2 marks]

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0 3 . 3 Explain why CHP stations must be located close to the users of the heat.

[1 mark]

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0 3 . 4 Explain how each of the following features of an industrial heat exchanger increases its efficiency.

[3 marks]

Long narrow copper pipes .....

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Counter-current flow .....

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Reduced flow rate .....

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0 3 . 5 Explain why a single large spherical tank loses less heat than several small cubic tanks holding the same total volume of hot fluid.

[2 marks]

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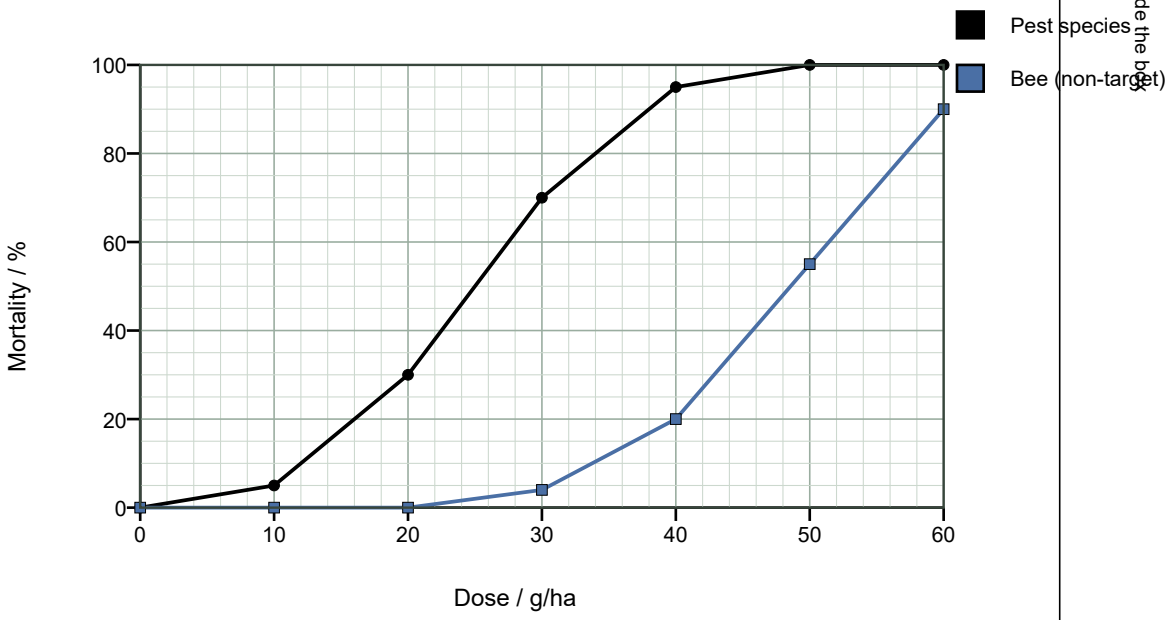
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0 4 Figure 1 shows how the percentage mortality of a pest species and of a non-target bee species change with the dose of an insecticide applied to a crop.

Figure 1



0 4 . 1 Use Figure 1 to estimate the LD_{50} of the insecticide for the pest species.

[1 mark]

Answer ..... g/ha

0 4 . 2 Define the term **MDNF** (maximum dose never fatal).

[1 mark]

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0 4 . 3 Use Figure 1 to explain why it may not be possible to kill all of the pest species without harming bees.

[3 marks]

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0 4 . 4 Suggest how the farmer could reduce the harm to bees without reducing the dose applied.

[2 marks]

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0 4 . 5 Explain why a pesticide with high specificity is less likely to cause environmental damage.

[3 marks]

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0 5 A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

0 5 . 1 Calculate the percentage water content of the fresh soil.

[2 marks]

Answer ____ %

0 5 . 2 Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer ____ %

0 5 . 3 Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 5 . 4 Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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0 5 . 5 Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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0 6 A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 6 . 1 Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

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0 6 . 2 Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using D = N(N-1) / Σn(n-1). Show your working.

[3 marks]

Answer

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0 6 . 3 The ungrazed meadow had D = 1.8 with the same number of species. Explain what this shows.

[2 marks]

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0 6 . 4 Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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0 6 . 5 Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

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0 7 Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

0 7 . 1 Complete Table 1 by explaining how each feature reduces oil pollution.

[5 marks]

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

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0 7 . 2 Explain how each of the following reduces the risk of tanker accidents.

[2 marks]

Offshore shipping routes

GPS / Automatic Identification System

0 7 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank.

[2 marks]

0 7 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park.

[1 mark]

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0 8 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 8 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 9 . 1 Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

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0 9 . 2 Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

[3 marks]

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0 9 . 3 Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

0 9 . 4 Describe **two** environmental impacts of solar PV.

[2 marks]

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1 0 . 1 Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by a **carbon footprint**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS